

A Machine Learning–Based Fall Risk Assessment Model for Inpatients

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Falls are one of the most common accidents among inpatients and may result in extended hospitalization and increased medical costs. Constructing a highly accurate fall prediction model could effectively reduce the rate of patient falls, further reducing unnecessary medical costs and patient injury. This study applied data mining techniques on a hospital's electronic medical records database comprising a nursing information system to construct inpatient-fall-prediction models for use during various stages of inpatient care. The inpatient data were collected from 15 inpatient wards. To develop timely and effective fall prediction models for inpatients, we retrieved the data of multiple-time assessment variables at four points during hospitalization. This study used various supervised machine learning algorithms to build classification models. Four supervised learning and two classifier ensemble techniques were selected for model development. The results indicated that Bagging+RF classifiers yielded optimal prediction performance at all four points during hospitalization. This study suggests that nursing personnel should be aware of patients' risk factors based on comprehensive fall risk assessment and provide patients with individualized fall prevention interventions to reduce inpatient fall rates.

KEY WORDS: Classification, Fall Risk assessment, Inpatient fall, Machine learning

The 2017 Centers for Disease Control and Prevention report states that one-third of older adults 65 years or older experience falls.¹ Falls are the most common factor and a major cause of casualties from accidental injuries

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This work was supported by Ditmanson Medical Foundation Chia-Yi Christian Hospital Research Program (grant number R109-37-2), and partially supported by the Ministry of Science and Technology (MOST) in Taiwan under grant nos. MOST 108-2636-H-194 -001 (Young Scholar Fellowship Program), and by the Center for Innovative Research on Aging Society (CIRAS) from The Featured Areas Research Center Program within the framework of the Higher Education Sprout Project by Ministry of Education (MOE) in Taiwan.

The authors have disclosed that they have no significant relationships with, or financial interest in, any commercial companies pertaining to this article.

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Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Web site (www.cinjournal.com).

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DOI: 10.1097/CIN.0000000000000727

among older adult inpatients. Studies have reported an inpatient fall rate of approximately 40%. Furthermore, 30% of fall patients experienced light to heavy injuries, ultimately resulting in increased medical costs and extended hospitalization days.^{2,3}

The causes of falls are complex and often include numerous intrinsic and extrinsic factors. Risk factors for falls include older age, female sex, a history of falls, specific comorbid conditions (eg, stroke, orthopedic diseases, anemia, postural hypotension, visual impairment, cognitive impairment, and urinary incontinence), surgery, impaired mobility, muscle weakness, medication (eg, benzodiazepines), and polypharmacy.^{4–9} Consequently, accurate assessment of the risk factors for falls and accessibility to fall prevention strategies can prevent falls effectively, further reducing unnecessary medical costs and patient injury.

Researchers have established several inpatient fall risk assessment scales with outstanding reliability and validity, such as the STRATIFY fall assessment scale,¹⁰ Hendrich II Fall Risk Model,¹¹ and Morse Fall Scale.¹² These scales serve as a reference for nurses providing nursing care plans. Because disease development is a dynamic process, patients' conditions may change with disease progression or administration of new drugs. Therefore, accurate identification of risk factors for falls becomes a major challenge in clinical care.

Because data mining and machine learning techniques have been used extensively in healthcare applications,^{13–17} studies have adopted data mining techniques to predict patient falls.^{6,15–20} Marschollek et al¹⁸ applied the C4.5 algorithm and logistic regression to construct fall prediction models and identified the risk factors for falls to be older age, low activities of daily living score, cognitive impairment, institutionalization, polypharmacy, and comorbidity. Another study used an artificial neural network to construct a prediction model for patient injury after falls and indicated that fall risk assessment score, antipsychosis medication, and diuretics were significant predictors.⁶ In addition to data mining techniques, some studies have applied text mining of the valuable unstructured data in electronic medical records (EMRs) and nursing records to predict risk factors for falls and accuracy. The results could provide guidelines and strategies for fall prevention and improving quality of care.^{21–24}

Most studies have emphasized identifying the risk factors for falls or predicting falls at a single point during

hospitalization.^{15,17,18} Hospitalization is a dynamic process, and inpatient conditions and the external environment may change over time, which means that the risk factors for inpatient falls may vary during the hospitalization period. To the best of our knowledge, studies rarely considered the changes in risk factors for falls at various points during hospitalization. A study used the Johns Hopkins Fall Risk Assessment Tool to assess patient status at three points of the hospitalization process, namely, upon hospital admission, before fall occurrence, and before being discharged. The study applied logistic regression and determined the risk factors for falls to be age, medications, patient care equipment, mobility, and cognition.¹⁹

This study applied data mining techniques on a hospital's EMR database comprising a nursing information system to construct inpatient-fall-prediction models for use during different stages of inpatient care. The risk factors for falls were generated using supervised machine learning technologies, including the decision tree (DT), Bayesian network (BN), support vector machine (SVM), and random forest (RF).

MATERIALS AND METHODS

Data Source and Preprocessing

This retrospective study retrieved inpatient information from the incident reporting system and the EMR database of a regional teaching hospital in Southern Taiwan. The inpatient data were collected from 15 inpatient wards, including the internal medicine department, surgical department, orthopedics department, and gynecology department, between January 2014 and May 2018. After exclusion of patients from the outpatient clinic and pediatric departments, a total of 108 940 cases remained, of which 256 patients were identified as inpatients who had falls. The study protocol was approved by the Institutional Review Board of Ditmanson Medical Foundation Chia-Yi Christian Hospital (CYCH-IRB no. 2018076). Written consent from the participants was deemed unnecessary because the dataset comprises anonymized secondary data for research purposes.

Variables

A total of 54 independent variables (IVs) were considered, which can be classified into nine categories, namely, demographics, admission diagnosis, self-care ability, physiological evaluation, surgical records, catheter records, focus charting, medication records, and fall scale score (Table 1).

The selected IVs can be divided into one-time and multiple-time assessment variables. One-time assessment variables, including demographics, admission diagnosis, self-care ability, and part of physiological evaluation (ie, consciousness, emotional assessment, visual impairment, malnutrition, sleep disorder, and having a prosthetic limb), can be measured on the day of admission. Multiple-time assessment variables,

including one of physiological evaluation (lower limb muscle strength), surgical records, catheter records, focus charting, medication records, and fall scale score, can be assessed during the hospitalization period.

To develop timely and effective fall prediction models for inpatients during their hospitalization periods, we retrieved the data of multiple-time assessment variables at four points:

- (1) *1d*: the first variable value collected within 24 hours of admission for both fall and nonfall groups.
- (2) *1st*: the first variable value collected after 24 hours of admission for both fall and nonfall groups.
- (3) *max*: the maximum variable value between admission and a fall event for the fall group, or the maximum variable value during the hospitalization period for the nonfall group.
- (4) *last*: the last variable value before a fall for the fall event group, or the last variable value during the hospitalization period for the nonfall group.

Data Analysis Instruments

All statistical analyses were performed using IBM SPSS Statistics version 22.0 (IBM Inc., Armonk, NY). Specifically, an independent *t* test and χ^2 test were conducted to examine the differences in all IVs between the fall and nonfall groups. Comparisons with $P < .05$ indicated statistically significant relationships. The CfsSubsetEval module with the greedy stepwise search algorithm in WEKA version 3.8.2 (Hamilton, New Zealand) was used for feature selection. This correlation-based feature selection method measures the correlations between the subsets of IVs and the dependent variable (DV). The optimal set of IVs contains variables that are highly correlated with the DV but uncorrelated with each other, which overcomes the multicollinearity issue within IVs.

For developing fall prediction models, numerous supervised learning techniques were selected, including DT, BN, SVM, and RF.^{25–30} Decision tree is a well-known classification technique, wherein each internal node in the tree represents a single IV, and each branch of an internal node represents a set of possible values of IV. For constructing a DT, the IV that can reduce node impurity the most is selected. At the end of the tree construction, each leaf node is assigned one class label as the prediction outcome.

Bayesian network is a graph-based probabilistic model based on Bayes' theorem. The basic principle of BN is that the conditional dependencies between any two variables can be represented as a directed acyclic graph. By using the values of IVs, the directed acyclic graph in BN can be used to calculate the probabilities of the presence of DV.

Support vector machine is one of the most effective supervised learning techniques for high-dimensional data; it adopts structural risk minimization to map IVs and a DV to a high-dimensional vector space. Subsequently, optimal

Table 1. Definition of Variables

Category	Variable Name	Definition	Type	Medical Record Type
Demographics	Sex	0 = Female 1 = Male	Nominal	One-time assessment record
	Age	Hospital date minus the date of birth	Numerical	
	Department	1 = Internal medicine 2 = Surgical 3 = Orthopedics 4 = Gynecology	Nominal	
Admission diagnosis	Dementia	Did this patient have these diagnoses? 0 = No 1 = Yes	Nominal	
	Depression			
	Diabetes mellitus			
	Cancer			
	Cardiovascular diseases			
	Neurologic diseases			
	Stroke			
	Musculoskeletal diseases			
	Eye diseases			
	Ear diseases			
	Anemia			
Orthostatic hypotension				
Self-care ability	Showering	1 = Complete dependence 2 = Partial dependence 3 = Self-care	Numerical	
	Changing clothes			
	Toileting			
	Preforming mobility functions			
	Using excretion function			
	Eating			
	Karnofsky Performance Scale	0 = Fully active, able to carry on all pre-disease performance without restriction 1 = Restricted in physically strenuous activity but ambulatory and able to carry out work of a light or sedentary nature, eg, light housework, office work 2 = Ambulatory and capable of all self-care but unable to carry out any work activities, up and about more than 50% of waking hours 3 = Capable of only limited self-care, confined to bed or chair more than 50% of waking hours 4 = Completely disabled, cannot carry on any self-care, totally confined to bed or chair		
	Barthel Index ^a	0–20 = total dependence 21–60 = severe dependence 61–90 = moderate dependence 91–99 = slight dependence		
	Consciousness	1 = Alert 2 = Anesthesia and not awake 3 = Drowsiness 4 = Stupor 5 = Comatose 6 = Other	Nominal	

(continues)

Table 1. Definition of Variables, Continued

Category	Variable Name	Definition	Type	Medical Record Type
	Emotional	1 = Calm 2 = Anxiety 3 = Depression 4 = Indifferent 5 = Excitement 6 = Sadness 7 = Fear 8 = Fear to death 9 = Intention to suicide 10 = Lost interest in daily activities 11 = Weepy 12 = Other	Nominal	
Physiological evaluation	Visual impairment (left/right)	0 = No	Nominal	Multiple-time assessment
	Malnutrition	1 = Yes		
	Sleep disorder			
	Prosthesis limb			
	Lower limb muscle strength	Refers to muscle strength 0 = No movement 1 = Flicker of movement only 2 = Movement possible when assisted by gravity or gravity is eliminated 3 = Movement possible against gravity but without imposed resistance 4 = Weak movement possible against gravity 5 = Normal movement against gravity and against imposed resistance	Nominal	
Surgical records	Surgery records	0 = No 1 = Yes	Nominal	
Catheter records	Peripheral intravenous catheter	0 = No 1 = Yes	Nominal	
Focus charting	Dizziness	0 = No	Nominal	
	Headache	1 = Yes		
	Diarrhea			
	Pollakiuria			
	Weakness			
	high-risk group of fall			
Medication records	Central nervous system drugs—anticonvulsants	Did this patient take these medications? 0 = No 1 = Yes	Nominal	
	Hematological agents—antithrombotic agents			
	Central nervous system drugs—general anesthetics			
	Skin and mucous membranes preparations—antipruritic and local anesthetics			
	central nervous system drugs—narcotic analgesics			
	Central nervous system drugs—sedatives and hypnotics			
	Central nervous system drugs—tranquilizers			
	Antihistamines and antiallergy agents			
	Electrolytic, caloric, and water balance—diuretics			

(continues)

Table 1. Definition of Variables, Continued

Category	Variable Name	Definition	Type	Medical Record Type
Fall scale score	Fall score	0–7 score	Numerical	
	Age	0 = <65 y 1 = ≥65 y		
	Consciousness	0 = Alert or coma 1 = Disturbance of consciousness		
	Fall history	0 = No history of fall 1 = History of fall		
	Walking ability	0 = Stable gaits or bed rest 1 = ≥unstable gaits or assistive devices		
	Toileting ability	0 = Self-care 1 = Toileting disability		
	Taking medications that have been associated with increased fall risk	0 = No 1 = Yes		
	Other (free text)	0 = No 1 = Yes		

^aThe total score is less than 60 points, which meets the National Health Insurance Home nursing care payment standard.

separating hyperplanes are identified to divide the samples into two or more categories. A class label of a new sample can be determined by mapping it into one of the subspaces projected by a set of hyperplanes.

Random forest is a DT-based classifier ensemble technique. In RF, both bootstrapping and random feature selection are performed on training data. Each DT is built from bootstrap samples with different number of IVs. The final predicted outcome is determined by a majority vote among multiple DTs.

Bagging and AdaBoost techniques were used to improve the prediction performance of the aforementioned single

classifiers.^{31–33} Bagging is an ensemble method that generates multiple bootstrap sample datasets from training datasets and builds base classifiers by using a single supervised learning technique. Given a new sample, the prediction output is estimated by averaging all the predictions made by the base classifiers. AdaBoost, one of the most well-known classifier ensembles, iteratively runs the selected learning algorithm and examines classification errors in the training dataset. If the current classifier misclassifies a training instance, the weight of this instance increases for the next round of learning. After a sequence of classifiers is constructed, AdaBoost adopts a weighted majority vote to aggregate prediction results.

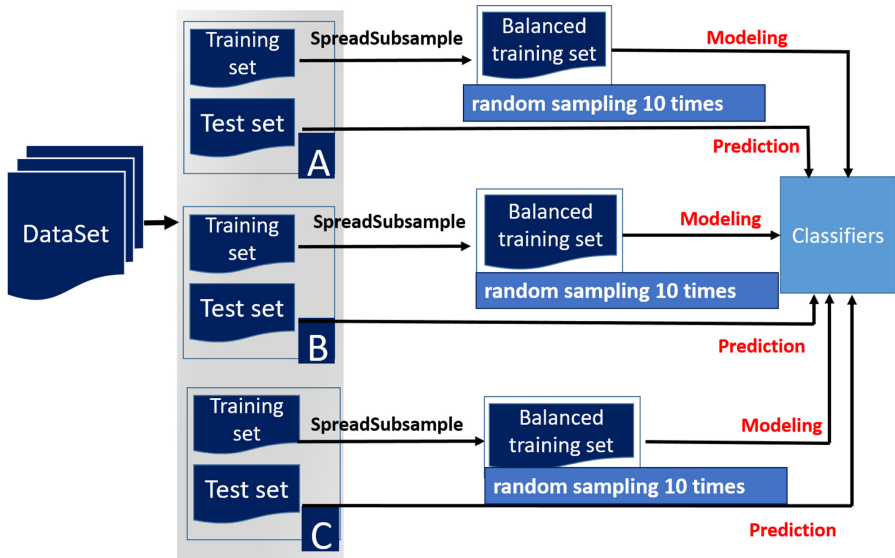


FIGURE 1. Experiment evaluation process.

Experimental Setup and Performance Measure

WEKA version 3.8.2 (Hamilton, New Zealand) was used to perform the supervised learning procedure. Specifically, DT, BN, SVM, and RF models were constructed using the J48, BayesNet, SMO, and RandomForest modules in WEKA, respectively. Default parameter settings in WEKA were adopted in all experimental evaluations.

The experimental evaluation process is illustrated in Figure 1. The RemoveFolds filter was applied in WEKA to divide the dataset into three folds, one of which was selected as the test set (1/3) and the remaining two as training sets (2/3). This filter was performed 10 times by varying the random seed value; thus, a total of 30 training and testing sets were generated. Because the proportions of the fall and nonfall patients considerably varied, the class imbalance problem may affect the learning performance during model development. Therefore, the SpreadSubsample module of WEKA was used to undersample the majority class (nonfall patients), and a dataset with identical numbers of fall and nonfall patients was generated. Additionally, to prevent bias incurred by the undersampling method, this study further adopted SpreadSubsample module 10 times for each training set to generate 10 balanced training sets.

The confusion matrix was used to evaluate prediction performance (Supplemental Digital Content 1, <http://links.lww.com/CIN/A86>), where true positive (TP) represents the number of inpatients who were correctly predicted to fall, true negative (TN) represents the number of inpatients who were correctly predicted to not fall, false positive (FP) represents the number of inpatients who were incorrectly predicted to fall, and false negative (FN) represents the number of inpatients who were incorrectly predicted to not fall. By use of the information summarized in the confusion matrix, three classification performance metrics, namely accuracy, sensitivity, and specificity, can be obtained using the following equations:

$$\text{Accuracy} = (TP + TN) / (TP + FP + FN + TN) \quad (1)$$

$$\text{Sensitivity} = TP / (TP + FN) \quad (2)$$

$$\text{Specificity} = TN / (TN + FP) \quad (3)$$

The area under the receiver operating characteristic curve (AUC) was included as an indicator of model performance, with larger AUCs indicating higher accuracy. Hosmer and Lemeshow³⁴ defined the prediction performance to be excellent if $AUC \geq 0.9$, good if $0.9 > AUC \geq 0.8$, and fair if $0.8 > AUC \geq 0.7$.

RESULTS

Descriptive Statistics

This study collected 108 940 inpatient cases between January 2014 and May 2018, which included 256 cases of falls. Falls

during hospitalization were set as the DV. Supplemental Digital Content 1 (<http://links.lww.com/CIN/A86>) presents the descriptive and inferential statistics of the patient demographic information. The average age of the patients was 60.2 (SD, 17.3) years, and 53.9% ($n = 58\,674$) were men. Most of the patients were admitted in the internal medicine department ($n = 58\,228$ [53.7%]). Nearly one-third of the patients had cancer ($n = 34\,132$, 31.3%).

In-hospital diagnosis revealed that 7.5% ($n = 8177$), 4.4% ($n = 4744$), and 4% ($n = 4371$) of the patients were diagnosed as having diabetes mellitus, stroke, and neurologic diseases, respectively, which were significantly associated with falls ($P < .05$). The patients' in-hospital self-care ability (ie, showering, changing clothes, toileting ability, performing mobility functions, using excretion functions, eating, and Karnofsky Performance Scale score) and sleeping disorders during the hospital admission period were significantly associated with fall risks ($P < .05$). Thus, patients' self-care dependency during the hospital admission period was related to fall risks. However, the patients' cognitive conditions, emotional conditions, visual impairment, malnutrition, and having prosthetic limbs during the hospital admission period did not have a significant association with fall risks (Supplemental Digital Content 1, <http://links.lww.com/CIN/A86>).

Experimental Results for the Prediction Model

Table 2 presents the prediction results of the selected classifiers, which indicated that the prediction performance of ensemble classifiers (Bagging and AdaBoost) was higher than that of single classifiers. The BN, Bagging+BN, and Bagging+RF classifiers generated optimal prediction results in the computation of the first patient evaluation datasets collected within 24 hours (1d), with an AUC value of 0.71. The RF, AdaBoost+RF, and Bagging+RF classifiers generated optimal results for the computation of the maximum value dataset (max) within multiple assessment patient records (AUC = 0.74).

In general, the Bagging+RF classifier generated the optimal prediction results for all four points during the inpatient hospitalizations: within 24 hours of hospital admission (1d, AUC = 0.71), after 24 hours of hospital admission (1st, AUC = 0.72), the maximum value dataset within multiple assessment patient records (max, AUC = 0.74), and the last recorded patient datasets (last, AUC = 0.76). The results indicated that predictions closer to the fall occurrence exhibited higher accuracy (Table 2).

DISCUSSION

The results revealed that the prediction results of the ensemble classifiers (ie, Bagging and AdaBoost algorithms) were more satisfactory than those of the single classifiers. The Bagging+RF model yielded optimal prediction results.

Table 2. Classification Performance Results

Classifier	Time	Accuracy	Sensitivity	Specificity	AUC
BN	1d	0.64	0.66	0.64	0.71
	1st	0.65	0.67	0.65	0.71
	Max	0.65	0.70	0.65	0.72
	Last	0.66	0.70	0.66	0.74
RF	1d	0.60	0.71	0.60	0.70
	1st	0.63	0.69	0.63	0.71
	Max	0.65	0.70	0.65	0.74
	Last	0.66	0.71	0.66	0.75
DT	1d	0.62	0.60	0.62	0.62
	1st	0.63	0.60	0.63	0.61
	max	0.65	0.61	0.65	0.63
	last	0.66	0.63	0.66	0.65
SVM	1d	0.64	0.64	0.64	0.64
	1st	0.65	0.63	0.65	0.64
	Max	0.67	0.64	0.67	0.66
	Last	0.68	0.65	0.68	0.67
AdaBoost+BN	1d	0.64	0.61	0.64	0.67
	1st	0.64	0.62	0.64	0.67
	Max	0.65	0.63	0.65	0.68
	Last	0.65	0.63	0.65	0.69
AdaBoost+RF	1d	0.60	0.70	0.60	0.70
	1st	0.63	0.68	0.63	0.71
	Max	0.66	0.70	0.66	0.74
	Last	0.66	0.71	0.66	0.75
AdaBoost+DT	1d	0.61	0.60	0.61	0.65
	1st	0.62	0.61	0.62	0.66
	Max	0.63	0.62	0.63	0.68
	Last	0.64	0.63	0.64	0.69
AdaBoost+SVM	1d	0.63	0.62	0.63	0.66
	1st	0.64	0.61	0.64	0.65
	Max	0.66	0.63	0.66	0.67
	Last	0.67	0.64	0.67	0.69
Bagging+BN	1d	0.65	0.66	0.65	0.71
	1st	0.65	0.67	0.65	0.71
	Max	0.65	0.70	0.65	0.72
	Last	0.67	0.70	0.67	0.74
Bagging+RF	1d	0.60	0.71	0.60	0.71
	1st	0.63	0.70	0.63	0.72
	Max	0.65	0.72	0.65	0.74
	Last	0.66	0.72	0.66	0.76
Bagging+DT	1d	0.62	0.65	0.62	0.68
	1st	0.63	0.65	0.63	0.69
	Max	0.65	0.66	0.65	0.71
	Last	0.65	0.68	0.65	0.72
Bagging+SVM	1d	0.67	0.60	0.67	0.69
	1st	0.68	0.59	0.68	0.69
	Max	0.70	0.61	0.70	0.71
	Last	0.71	0.62	0.71	0.72

Moreover, predictions closer to the fall occurrence indicated higher accuracy. Therefore, this classification model is applicable for making continuous predictions throughout the inpatient care process.

The feature selection function (GainRatioAttributeEval) of the WEKA software program revealed key variables influencing fall risks during inpatient care, the importance of the IVs including: fall history, walking ability, depression, stroke, low lower limb muscle strength, reduced self-care abilities (ie, using excretion function), and use of central nervous system drugs—sedatives and hypnotics, and classification in the high-risk group for fall incidents (Figure 2). Because of the long bedridden periods, patients may experience decreased lower limb muscle strength, which influences their gait stability and thereby increases fall risks. Jeon et al³⁵ investigated fall risks in 110 patients in nursing homes older than 60 years and revealed that patients' decreased lower limb muscle strength, unstable gait, poor balance, and fear of falling significantly increased the risk of falls. Therefore, further analysis is required to determine whether decreased lower limb muscle strength or unsteady gait is a fall risk factor.

Moreover, patients experiencing insufficient muscle strength when walking to the restroom or dizziness when standing up after using the restroom may experience falls. This result is consistent with that of Weil,⁷ who conducted a literature review that revealed that most falls occurred at night and mainly at the bedside (61.0%) or in the restrooms (11.5%). Therefore, nursing personnel should constantly monitor the changes in patients' lower limb muscle strength, adopt early intervention measures, and provide health education to patients to prevent falls caused by patients' insufficient muscle strength.

We also determined that the use of central nervous system drugs—sedatives and hypnotics—is one of the key fall risk factors. This result is consistent with that reported by Hayakawa et al,³⁶ who demonstrated that hypnotics and psychoactive drugs exert similar effects. Their study indicated that fall occurrence in women is associated with cognitive dysfunction, surgical history, and sedative-hypnotic drug use. Among men, fall occurrence is associated with the use of psychoactive drugs. Lee et al⁶ investigated the fall information reported by an educational and regional hospital in Southern Taiwan and adopted machine learning to predict

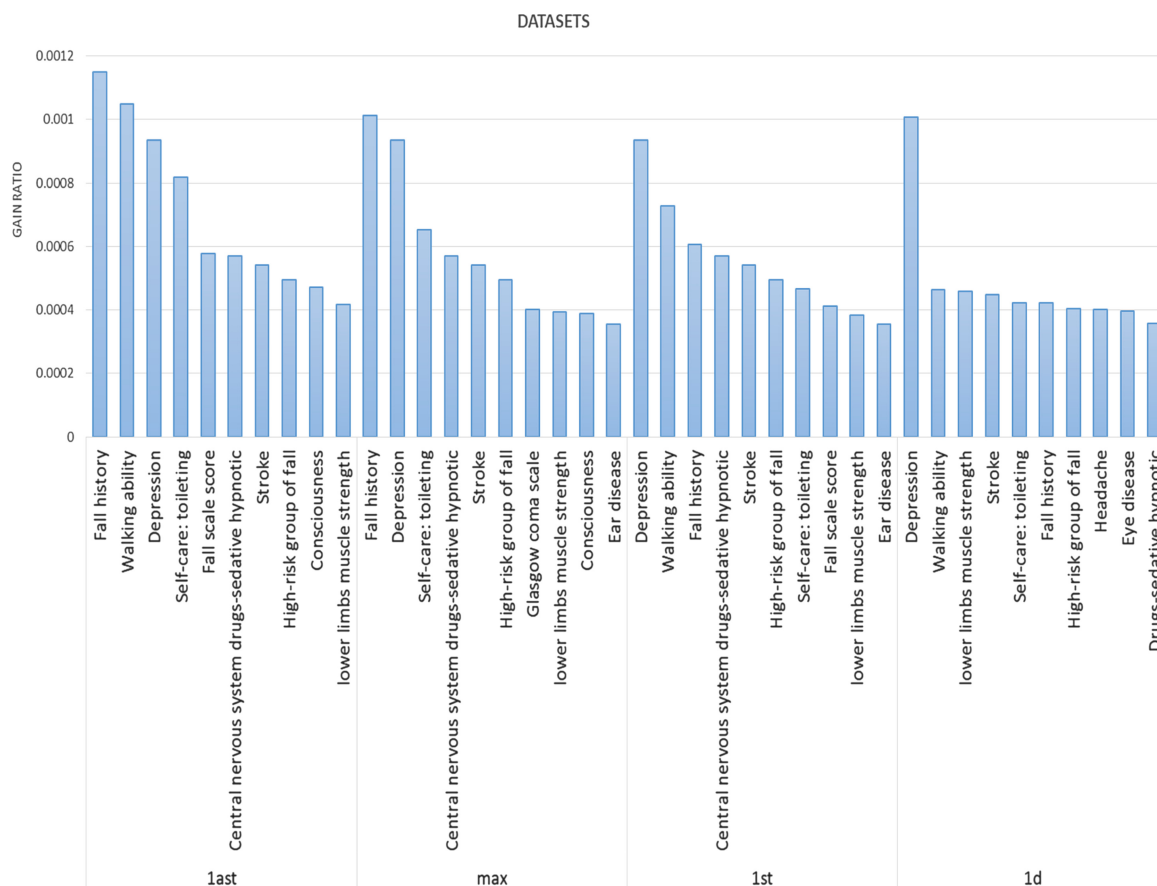


FIGURE 2. Importance of the variables.

the degree of injury caused by falls. The results indicated that the use of psychoactive drugs and diuretics could predict the degree of injury after fall incidents.⁶

Intervention measures for fall prevention should be established in clinical nursing fields, including educating primary caregivers on relevant health measures and precautions to be taken when accompanying patients leaving their bed to the restroom.

CONCLUSION

Falls are one of the most common accidents among inpatients. The degree of fall injuries increases with the patient's age. Medical institutions should establish individual fall assessment scales with sufficient reliability and validity to screen for risk factors for falls and should establish early intervention measures. As the inpatient care period is dynamic; risk factors for falls may vary according to changing patient conditions. To determine the risk factors during various inpatient care stages and establish adequate intervention strategies for caregiving personnel, this study developed prediction models according to various patient characteristics and the multiple-time assessment variables at four points during hospitalization. We suggest that nurses be aware of patients' risk factors for falls and provide patients with individualized fall prevention interventions to reduce inpatient fall rates and severity of injury. As a consequence, we hope to reduce the workload of medical personnel and prevent unnecessary extended hospitalization or medical costs for patients.

This study has some limitations. Because the established prediction model used information collected from a single hospital, the results might not be applicable to patients in other hospitals. In the future, researchers could validate the model through cross-hospital cooperation. In addition, this study mainly collected fall data during the inpatient care period from the hospital incident reporting system, which contains data actively reported by nursing personnel. Therefore, further clarification of whether clinical reports are missing is required. Finally, recent studies have revealed that text mining is suitable for capturing usable data from unstructured nursing records. Future research may consider adopting text mining to enhance the accuracy of fall risk predictions, which can be used as a guide by nursing personnel to establish effective fall prevention interventions and reduce fall and injury rates.

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